The Department of Textile Engineering and Design operates under the Faculty of Engineering and Architecture and is fully accredited by the Pakistan Engineering Council. Its core focus encompasses the full manufacturing lifecycle of textiles, transforming raw fibers into yarn, spinning yarn into fabric, and executing garment, apparel, and home textile designs, while also including the processing of wool, cotton, and manmade fibers, technical textiles, color science, and environmental sustainability in manufacturing. The department was launched in 2004 as a pioneer engineering program at the Takatu Campus with an initial intake of 20 students, strategically positioned because Balochistan is rich in raw wool production, and it introduced its postgraduate MS program in 2015. Graduates pursue career pathways working as technical experts, supervisors, and managers across textile processing plants, quality control units, research labs, product development divisions, and corporate management, with the field also branching into specialized biomedical applications such as medical filters and artificial arteries. The vision of the department is to be recognized internationally for its quality research, educational activities, and its relationship with the environment, aiming for its graduates to be acknowledged for the strength of their technical and practical knowledge, their ability to work as a team, and their vocation of service to society. Its mission is to provide professional training, promoting social responsibility, values, and a progressive research culture among the students of Textile Engineering, making them capable of providing solutions to complex engineering problems, preserving the environment, and contributing to the sustainable development of Pakistan and the society.

For the BS Textile Engineering Engineering Track, admission requirements mandate an academic background of F.Sc Pre-Engineering or ICS with Mathematics and Physics from a recognized board or an equivalent qualification with at least 60% marks, or a Diploma of Associate Engineer in a relevant discipline with a minimum of 60% marks, and success in the NTS Test is required. The degree requirements entail completing a total of 136 credit hours across 50 courses, maintaining a minimum academic standing CGPA of 2.0 or higher. The Program Educational Objectives aim to implement real-world applications making use of fundamental engineering knowledge, designs, and managerial skills gained during the program, to generate innovative and sustainable solutions to complex engineering problems contributing to the textile industry, and to promote graduates through communication and interpersonal skills, career management strategies, and the ability to deal with technological, socio-economical, environmental, and ethical challenges.

The curriculum plan details the program schema by semester, noting that credit hour listings follow a theory and lab format. In the first semester, students take Linear Algebra and Geometry or Applied Chemistry with no prerequisites for three and zero or three and one credit hours respectively, Applied Physics or Textile Raw Materials-I with no prerequisites for three and one or three and zero credit hours respectively, and Functional English or Introduction to Textile Engineering with no prerequisites for two and zero credit hours each. The second semester consists of Calculus-I or Engineering Drawing and Graphics with no prerequisites for two and zero or zero and one credit hours respectively, Engineering Mechanics or Textile Raw Materials-II with no prerequisites for two and one or three and zero credit hours respectively, Islamic Studies or Introduction to Computing with no prerequisites for two and zero or one and one credit hours respectively, and Communication Skills or Thermodynamics with no prerequisites for two and zero or two and one credit hours respectively.

The third semester involves Calculus-II or Fibre Science with no prerequisites for two and zero or three and one credit hours respectively, Machine Design or Pakistan Studies with no prerequisites for two and zero credit hours each, and Spinning Preparatory Process or Fabric Manufacturing-I with no prerequisites for two and one credit hours each. The fourth semester covers Statistics and Probability or Electrical and Electronic Engineering with no prerequisites for three and zero or two and one credit hours respectively, Professional Ethics with no prerequisite for two and zero credit hours, and Yarn Production Process or Fabric Manufacturing-II with no prerequisites for three and one credit hours each.

In the fifth semester, students take Organizational Behavior or Technical Writing and Presentation Skills, with the former having no prerequisite and the latter requiring English-I, for two and zero or three and zero credit hours respectively, alongside Spinning Planning and Production or Weave Designs with no prerequisites for two and one or three and one credit hours respectively, and Pre-treatment in Textile Wet Processing or Garment Fundamentals with no prerequisites for two and one credit hours each. The sixth semester requires Operations and Project Management or Process Systems with no prerequisites for two and zero or two and one credit hours respectively, Textile Dyes and Dyeing with a prerequisite of Applied Chemistry for two and one credit hours, Anthropometry and Clothing Construction or Technical Textiles with no prerequisites for two and one or two and zero credit hours respectively, Textile Utilities and Services or Color Science with no prerequisites for two and zero credit hours each, and a mandatory non-credit Industry Internship lasting four to six weeks.

The seventh semester includes Engineering Economics or Merchandizing with no prerequisites for two and zero credit hours each, Design Project-I or Computer Application in Textiles, where the latter requires Computing, for zero and three or zero and two credit hours respectively, Mechanism of Garment Machines or Sewn Production Engineering with no prerequisites for two and one credit hours each, and Denim Technology with no prerequisite for two and zero credit hours. The eighth semester concludes the engineering track with Entrepreneurship or Textile Testing and Quality Control with no prerequisites for two and zero or three and one credit hours respectively, Environmental Issues and Control with no prerequisite for two and one credit hours, and Textile Printing and Finishing or Design Project-II with no prerequisites for three and one or zero and three credit hours respectively.

For the BS Fashion and Textile Design Non-Engineering Track, admission requirements state that an academic background of Intermediate certification such as F.Sc, ICS, or F.A. from any recognized board or an equivalent qualification with at least 45% marks is needed, and applicants must pass both the central NTS Test and a localized Free Hand Drawing Test. The degree requirements for this track mandate completing a total of 136 credit hours across 64 total courses while maintaining a minimum academic standing CGPA of 2.0 or higher.